High-Order Co-normal Flux Correction

Variable Robin Boundary Conditions in Dense Jacobi-Simplex HPS

Technical derivation and implementation note - Sachin Natesh

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Abstract

The existing hierarchical Poincaré–Steklov (HPS) skeleton carries degree-n face traces and a degree-n *tau-stabilized numerical co-normal flux*. For polynomial diffusion of degree $p_A \geq 2$, the physical co-normal flux of a degree-n volume solution can have degree greater than n. A spatially varying Robin coefficient β can couple these omitted high modes back into the degree-n boundary equations. We derive a correction that observes the physical high flux on true exterior faces, propagates its affine dependence through existing HPS interface reconstruction maps, and modifies only the root Robin operator and right-hand side. Neither the degree-n interface skeleton nor its tau-stabilized merge equations is enlarged. The current intended implementation is **Dense-only**; DenseSparse, MatrixFree, and Verify remain outside this extension.

1 Equation, approximation spaces, and the missing information

Consider a conforming affine simplicial mesh in dimension D, and the divergence-form equation

$$-\nabla\cdot\big(A(x)\nabla u\big) + b(x)\cdot\nabla u + c(x)u = f(x), \tag{1}$$

with Robin data on the physical exterior boundary $\partial\Omega$:

$$\alpha(x)u(x) + \beta(x)q(x) = g(x), \qquad q = n_{\text{out}}^{T}A(x)\nabla u(x). \tag{2}$$

On each element K, approximate u by degree-n volume Jacobi polynomials. The κ-weighted volume basis and its residual family $\kappa_{\text{res}} = \kappa + \mathbf{2}$ follow the existing implementation convention. The divergence-form coefficient adapter forms

$$\widetilde{A}_{rs} = -A_{rs}, \qquad \widetilde{b}_s = b_s - \sum_{r=1}^{D}\partial_{x_r}A_{rs}, \qquad \widetilde{c} = c, \tag{3}$$

then promotes the coefficients to κ_{res}. The expanded interior operator is unchanged. As the existing leaf convention is $L\mathbf{c} = -\mathbf{f}^{\text{int}}$, input physical source moments use $\mathbf{f}^{\text{int}} = -\mathbf{f}_{\text{phys}}$. This sign convention does not alter any of the flux formulas below.

Every face uses the *same canonical* degree-n Jacobi face basis, with face family given by the first D entries of the volume parameter vector. Put

$$V_L = \Pi_n^{D-1}, \qquad k_L = \dim V_L. \tag{4}$$

Let $\mathcal{P}_L$ denote the weighted face-moment projection onto V_L. For a degree-p_A diffusion tensor,

$$u|_K \in \Pi_n^D \quad\Longrightarrow\quad q|_f \in \Pi_{p_H}^{D-1}, \qquad p_H = n + p_A - 1 \tag{5}$$

(as an upper bound). We split the face coefficient vector of the *physical* flux into low degrees $0{:}n$ and high degrees $n+1{:}p_H$:

$$\mathbf{q}_L = \mathcal{P}_L q, \qquad \mathbf{q}_H = \mathcal{P}_H q, \quad V_H = \Pi_{p_H}^{D-1} \ominus \Pi_n^{D-1}. \tag{6}$$

Here $\ominus$ means the orthogonal difference in the common face basis. These are coefficient vectors of weighted moments; physical face measure factors are included consistently in trace/flux maps. If $p_A \le 1$, the high space is empty. For $D = 1$, a face is a point and also has no high modes.

2 The ordinary tau-stabilized HPS leaf

Let $\mathbf{c}_K$ be the element's volume modal coefficients and $\boldsymbol{\lambda}_K$ its stacked degree-n trace degrees of freedom. Denote the interior residual, trace, and *low physical co-normal flux* maps by

$$L_K \mathbf{c}_K, \qquad T_K \mathbf{c}_K, \qquad F_{A,K} \mathbf{c}_K = \mathbf{q}_{L,K}. \tag{7}$$

The tau-stabilized least-squares system has the schematic form

$$\mathcal{A}_{\tau,K} \mathbf{c}_K \simeq \begin{bmatrix} -s_{L,K} \mathbf{f}_K^{\mathrm{int}} \\ \mathcal{T}_K^{1/2} \boldsymbol{\lambda}_K \end{bmatrix}, \qquad \mathcal{A}_{\tau,K} = \begin{bmatrix} s_{L,K} L_K \\ \mathcal{T}_K^{1/2} T_K \end{bmatrix}, \tag{8}$$

where $\mathcal{T}_K$ is the existing (facewise degree-n) tau penalty and $\simeq$ denotes the least-squares solve. The response maps already produced by DenseQR are

$$\mathbf{c}_K = U_{\lambda,K} \boldsymbol{\lambda}_K + U_{f,K} \mathbf{f}_K^{\mathrm{int}}, \qquad \begin{aligned} U_{\lambda,K} &= \mathcal{A}_{\tau,K}^{\dagger} \begin{bmatrix} 0 \\ \mathcal{T}_K^{1/2} \end{bmatrix}, \\ U_{f,K} &= \mathcal{A}_{\tau,K}^{\dagger} \begin{bmatrix} -s_{L,K} I \\ 0 \end{bmatrix}. \end{aligned} \tag{9}$$

The skeleton uses the *augmented* degree-n co-normal flux

$$\widehat{\boldsymbol{\mu}}_K = F_{A,K} \mathbf{c}_K - \mathcal{T}_K (T_K \mathbf{c}_K - \boldsymbol{\lambda}_K) = S_K \boldsymbol{\lambda}_K + \mathbf{b}_K, \tag{10}$$

with

$$\boxed{S_K = (F_{A,K} - \mathcal{T}_K T_K) U_{\lambda,K} + \mathcal{T}_K, \qquad G_{f,K} = (F_{A,K} - \mathcal{T}_K T_K) U_{f,K}, \qquad \mathbf{b}_K = G_{f,K} \mathbf{f}_K^{\mathrm{int}}.} \tag{11}$$

Here $\mathbf{b}_K$ is a *source-dependent numerical-flux offset*, not the PDE advection coefficient $b(x)$. The HPS tree stores S_K once, updates $\mathbf{b}_K$ for each source, and reconstructs $\mathbf{c}_K$ after the downward trace pass. Changing $F_{A,K}$ does not change $U_{\lambda,K}$ or $U_{f,K}$.

3 HPS interface elimination and the source upward pass

Consider a merge of child subdomains A and B. Partition each child's degree-n boundary trace into exterior-to-the-merge blocks E and the common interface block I. Let $\boldsymbol{\gamma}$ be the common interface trace. The children satisfy

$$\begin{bmatrix} \widehat{\boldsymbol{\mu}}_{A,E} \\ \widehat{\boldsymbol{\mu}}_{A,I} \end{bmatrix} = \begin{bmatrix} S_{A,EE} & S_{A,EI} \\ S_{A,IE} & S_{A,II} \end{bmatrix} \begin{bmatrix} \boldsymbol{\lambda}_{AE} \\ \boldsymbol{\gamma} \end{bmatrix} + \begin{bmatrix} \mathbf{b}_{A,E} \\ \mathbf{b}_{A,I} \end{bmatrix}, \tag{12}$$

with an analogous equation for B. The interface conditions are *continuity of* $\boldsymbol{\gamma}$ and conservation of the *augmented* numerical flux,

$$\widehat{\boldsymbol{\mu}}_{A,I} + \widehat{\boldsymbol{\mu}}_{B,I} = 0. \tag{13}$$

Define the interface matrix $K_I = S_{A,II} + S_{B,II}$. With outward co-normals on each child, the eliminated trace is

$$\begin{aligned} \boldsymbol{\gamma} &= R_A \boldsymbol{\lambda}_{AE} + R_B \boldsymbol{\lambda}_{BE} + \mathbf{r}, \\ R_A = -K_I^{-1} S_{A,IE}, \qquad R_B &= -K_I^{-1} S_{B,IE}, \qquad \mathbf{r} = -K_I^{-1}(\mathbf{b}_{A,I} + \mathbf{b}_{B,I}). \end{aligned} \tag{14}$$

The parent face ordering is $[AE, BE]$. Its DtN matrix is the usual Schur-complement merge of child matrices; in particular, its source offset is

$$\boxed{\mathbf{b}_{P,AE} = \mathbf{b}_{A,E} + S_{A,EI}\mathbf{r}, \qquad \mathbf{b}_{P,BE} = \mathbf{b}_{B,E} + S_{B,EI}\mathbf{r}.} \tag{15}$$

The stored R_A, R_B depend only on the homogeneous child DtN matrices. The source pass refreshes $\mathbf{b}_A, \mathbf{b}_B, \mathbf{r}, \mathbf{b}_P$ without rebuilding the merge; the implementation also retains $R_f = -K_I^{-1}$, so $\mathbf{r} = R_f(\mathbf{b}_{A,I} + \mathbf{b}_{B,I})$. At the root, the ordinary HPS relation is

$$\widehat{\boldsymbol{\mu}}_{\text{root}} = S_{\text{root}} \boldsymbol{\lambda}_{\text{root}} + \mathbf{b}_{\text{root}}. \tag{16}$$

For *constant* α, β, solve $(\alpha I + \beta S_{\text{root}})\boldsymbol{\lambda}_{\text{root}} = \mathbf{g} - \beta \mathbf{b}_{\text{root}}$, then reconstruct $\boldsymbol{\gamma}$ downward and each $\mathbf{c}_K$ through (9). Nothing in this standard HPS procedure requires high-order skeleton unknowns.

4 Why variable β needs a high-flux correction

The degree-n root Robin equations must represent the *projected product* in (2), rather than a product of independently truncated data. The intended discrete rule (formulation B) is

$$\mathcal{P}_L\big(\alpha\, \boldsymbol{\lambda} + \beta\, \widehat{q}_{\text{full}}\big) = \mathbf{g}, \qquad \widehat{q}_{\text{full}} = q_{\text{physical}} - \underbrace{\mathcal{T}(T\mathbf{c} - \boldsymbol{\lambda})}_{\text{degree} \le n}. \tag{17}$$

The penalty is already in the low numerical-flux representation (10). It contributes *no high modes*, because the current tau weight is constant on each affine face and the trace discrepancy lies in Π_n^{D-1}. Thus the full numerical flux is represented exactly, up to its available polynomial degree, by the pair

$$(\widehat{\boldsymbol{\mu}}_L, \mathbf{q}_H) \equiv (S_{\text{root}} \boldsymbol{\lambda}_{\text{root}} + \mathbf{b}_{\text{root}},\ \mathcal{P}_H q_{\text{physical}}). \tag{18}$$

For a face polynomial β, let $M_\beta^{L \leftarrow L}$ and $M_\beta^{L \leftarrow H}$ be the *same weighted face multiplication operator*, with output projected to V_L and input restricted to low or high modes. If p_H is the input high cutoff, these are adjacent column blocks of

$$M_\beta^{n \leftarrow p_H} : \Pi_{p_H}^{D-1} \longrightarrow \Pi_n^{D-1}, \qquad (M_\beta)_{ij} = \int_{\widehat{f}} \psi_i\, \beta\, \psi_j\, w_{\kappa_{\text{face}}}. \tag{19}$$

The face multiplication plan and the flux moment coefficients must use the same canonical face normalization. Keeping only $M_\beta^{L \leftarrow L} \widehat{\boldsymbol{\mu}}_L$ discards the nonzero term $M_\beta^{L \leftarrow H} \mathbf{q}_H$.

There is a useful degree restriction. If $\deg \beta = p_\beta$, orthogonality gives

$$M_\beta^{L \leftarrow H} \psi_j = 0 \quad \text{for high input modes of degree } \deg \psi_j > n + p_\beta. \tag{20}$$

Therefore only degrees

$$n + 1 \le d \le p_{\text{obs}}, \qquad p_{\text{obs}} = n + \min(p_A - 1, p_\beta) \tag{21}$$

need be retained for a specified β. No high correction is needed for constant β, $p_A \le 1$, or $D = 1$. Retaining all degrees through p_H instead is simpler and independent of subsequent updates to β.

5 High-order observation at a leaf

For a globally exterior face $f \subset \partial K \cap \partial\Omega$, construct the rectangular *physical* flux moment map

$$Q^f_{H,K}\mathbf{c}_K = \mathcal{P}^f_H(n_f^T A\nabla u_{\mathbf{c}_K}), \tag{22}$$

where H selects degrees $n+1{:}p_H$ (or the smaller relevant range (21)). The map is a high-degree companion of the existing tested co-normal face map. In intrinsic canonical face coefficients, it is assembled by

$$Q^f_{H,K} = \mathrm{rows}_H\left[J_f \sum_{s=1}^{D} M^{p_H\leftarrow n}_{w^f_s} G^f_s\right], \qquad w^f_s = \sum_{r=1}^{D}(n_f)_r A_{rs}|_f, \tag{23}$$

with G^f_s the intrinsic face coefficients of $\partial_{x_s} u$ and J_f the physical face scale. Equivalently, use the already scaled normal $J_f n_f$ inside w^f_s, but *not both*: physical face measure is applied once. The rectangular multiplication must be integrated with sufficient face quadrature to resolve the polynomial triple products; it must not first truncate $A\nabla u$ to degree n.

Stack $Q^f_{H,K}$ over *only the faces on the true exterior boundary*. Call this compact row-stacked operator $Q_{H,K}$. Using (9),

$$\boxed{\mathbf{q}_{H,K} = H_K\boldsymbol{\lambda}_K + \mathbf{z}_K, \qquad H_K = Q_{H,K}U_{\lambda,K}, \qquad \mathbf{z}_K = Q_{H,K}U_{f,K}\mathbf{f}^{\mathrm{int}}_K.} \tag{24}$$

The *rows* of H_K correspond only to selected global boundary faces, but its *columns* include *all* low-order faces of K, including faces that will be eliminated as interfaces. This distinction is essential.

6 Propagate observations through the existing merge tree

Suppose children A, B carry high observations on some true exterior faces:

$$\mathbf{q}_{H,A} = H_A\boldsymbol{\lambda}_A + \mathbf{z}_A, \qquad \mathbf{q}_{H,B} = H_B\boldsymbol{\lambda}_B + \mathbf{z}_B. \tag{25}$$

Partition the *columns* of these maps into each child's exterior-to-merge and interface trace blocks:

$$\mathbf{q}_{H,A} = H_{A,E}\boldsymbol{\lambda}_{AE} + H_{A,I}\boldsymbol{\gamma} + \mathbf{z}_A, \quad \mathbf{q}_{H,B} = H_{B,E}\boldsymbol{\lambda}_{BE} + H_{B,I}\boldsymbol{\gamma} + \mathbf{z}_B. \tag{26}$$

Substitute the *already available* interface reconstruction (14). In parent column ordering $[\boldsymbol{\lambda}_{AE}; \boldsymbol{\lambda}_{BE}]$, this yields the explicit upward formulas

$$\boxed{H_P = \begin{bmatrix} H_{A,E} + H_{A,I}R_A & H_{A,I}R_B \\ H_{B,I}R_A & H_{B,E} + H_{B,I}R_B \end{bmatrix}, \qquad \mathbf{z}_P = \begin{bmatrix} \mathbf{z}_A + H_{A,I}\mathbf{r} \\ \mathbf{z}_B + H_{B,I}\mathbf{r} \end{bmatrix}.} \tag{27}$$

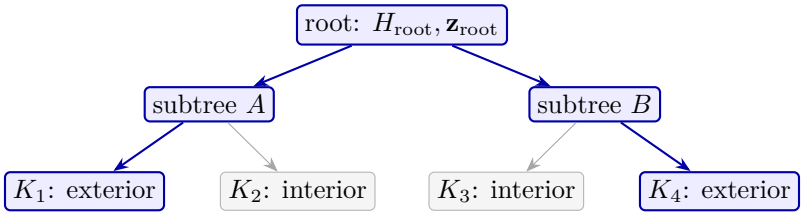
Rows are concatenated in the chosen ordering of the selected exterior faces; zero-row children simply contribute nothing. Observe that H_P uses only the source-independent R_A, R_B, while $\mathbf{z}_P$ uses the current source-dependent interface offset $\mathbf{r}$. There are *no additional high-order interface continuity or conservation conditions*: high modes are observations, not new trace unknowns.

Which branches need this propagation?

Let $\mathcal{L}_\partial$ be the set of leaf elements incident to a *true global exterior face*. Define the *active ancestor closure*

$$\mathcal{A} = \bigcup_{K\in\mathcal{L}_\partial} \mathrm{path}(K, \mathrm{root}). \tag{28}$$

Build high observations on the selected faces of these leaves only, and form $(H_N, \mathbf{z}_N)$ only for nodes $N \in \mathcal{A}$. At a merge, propagate any active child; an inactive child has no observation rows. All ordinary S_N, $\mathbf{b}_N$, $R_A, R_B, \mathbf{r}$ are still computed on the *whole* HPS tree.



Schematic: only highlighted leaf-to-root paths carry high-flux observations. Nevertheless, a source supported in an inactive interior subtree changes its ordinary $\mathbf{b}$, hence the interface offset $\mathbf{r}$ encountered on an active path, and therefore $\mathbf{z}_{\text{root}}$. "Inactive" means *no high-observation rows*, not "irrelevant to the physical solution."

Consequently the final root observation is

$$\boxed{\mathbf{q}_{H,\text{root}} = H_{\text{root}}\boldsymbol{\lambda}_{\text{root}} + \mathbf{z}_{\text{root}}, \qquad \text{rows indexed only by global exterior high-flux modes.}} \tag{29}$$

For a fixed coefficient field, H_{root} is reusable across many right-hand sides. For each source update, first refresh ordinary child offsets and $\mathbf{r}$ in the usual upward pass, then refresh $\mathbf{z}_N$ on active branches using (27); no new local least-squares solves are required.

7 Corrected root Robin equation

Let M_α^{LL} be the ordinary low-to-low face multiplication by α; let M_β^{LL} and M_β^{LH} be the two column blocks of the rectangular degree-n output multiplication by β. All are block diagonal over *global exterior faces*, ordered consistently with S_{root}, $\mathbf{b}_{\text{root}}$, and $(H_{\text{root}}, \mathbf{z}_{\text{root}})$. Projecting (17) gives

$$\mathbf{g} = M_\alpha^{LL}\boldsymbol{\lambda}_{\text{root}} + M_\beta^{LL}(S_{\text{root}}\boldsymbol{\lambda}_{\text{root}} + \mathbf{b}_{\text{root}}) + M_\beta^{LH}(H_{\text{root}}\boldsymbol{\lambda}_{\text{root}} + \mathbf{z}_{\text{root}}). \tag{30}$$

Thus *only the root Robin matrix and its right-hand side change*:

$$\boxed{\begin{aligned} R_{\text{root}} &= M_\alpha^{LL} + M_\beta^{LL}S_{\text{root}} + M_\beta^{LH}H_{\text{root}}, \\ \mathbf{h}_{\text{root}} &= \mathbf{g} - M_\beta^{LL}\mathbf{b}_{\text{root}} - M_\beta^{LH}\mathbf{z}_{\text{root}}, \\ R_{\text{root}}\boldsymbol{\lambda}_{\text{root}} &= \mathbf{h}_{\text{root}}. \end{aligned}} \tag{31}$$

After solving, the standard *unchanged* downward pass reconstructs interfaces by (14) and volume solutions by (9). The high flux is not used to modify merges, interface conservation, tau penalties, or the degree-n trace skeleton.

Sanity checks and implementation invariants

- **Constant β:** orthogonality gives $M_\beta^{LH} = 0$, recovering the validated constant-Robin equation, with no need to construct H_{root}.
- **Low-degree diffusion:** if $p_A \le 1$, the physical co-normal flux has no high degree beyond n. Again the correction vanishes.
- **Zero source:** with $\mathbf{f}^{\text{int}} = 0$, all $\mathbf{b}_N$, $\mathbf{r}$, and $\mathbf{z}_N$ vanish. The new term is then purely $M_\beta^{LH}H_{\text{root}}$.

- **Nonzero source:** retain the $M_\beta^{LH}\mathbf{z}_{\text{root}}$ term. Neglecting it breaks the affine particular-solution contribution even if the operator correction is correct.
- **Full numerical flux:** multiply the *low augmented flux* and the *high physical flux*; do not apply β only after truncating the physical flux, and do not introduce an additional high-order tau penalty.
- **Face conventions:** use the common canonical face κ, correct orientation permutations, and exactly one physical face scaling. Ensure both M_β^{LL} and M_β^{LH} derive from one correctly quadrature-resolved rectangular multiplication operator.
- **Backend scope:** implement and validate this on the `Dense` leaf path only. Reject unsupported `DenseSparse`, `MatrixFree`, and `Verify` divergence/high-flux configurations in the public interface.

Minimal implementation sequence. (1) Assemble rectangular high physical co-normal face maps and compare with independent weighted face quadrature; (2) form leaf H_K and $\mathbf{z}_K$ on global boundary faces; (3) reuse each merge's existing R_A, R_B and source offset $\mathbf{r}$ to propagate (27) only on $\mathcal{A}$; (4) assemble $M_\beta^{n\leftarrow p_H}$, extract its low/high column blocks, and solve (31); (5) validate variable Robin polynomial manufactured solutions, then nonhomogeneous and convergence cases. These steps preserve the current Dense HPS interfaces and source-reuse architecture.